Stomatal traits across cytotypes — short summary

Preliminary results from automated analysis of micrographs

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22 October 2026

Summary

We automatically detected and measured stomata in **3,253 micrographs** from **540 plants** (5 populations; four cytotypes; both leaf sides), giving **105,354 detected stomata**. Key results:

- **Cell size follows ploidy, but not linearly.** Synthetic neotetraploids (`neo_4x`) have the largest stomata (23.9 μm abaxial, +27% vs. `nat_2x`), while established natural tetraploids (`nat_4x`) are intermediate (20.6 μm, +10%). Diploids `ory_2x` and `nat_2x`: 18.5 and 18.7 μm.
- **Density.** Abaxial density is 254 mm^{-2} in `nat_2x`, 237 in `nat_4x` (-7%), 164 in `neo_4x` (-35%) and 297 in `ory_2x`.
- **Theoretical conductance.** Abaxial $g_{s,\max}$ (mol m^{-2} s^{-1}): `neo_4x` 0.76, `nat_4x` 0.92, `nat_2x` 0.91, `ory_2x` 1.03. See Table 2 for which differences are statistically supported.
- **Leaf sides.** All cytotypes are amphistomatous; see Table 1 for abaxial vs. adaxial values.

Methods in brief

Image analysis. A YOLO11 instance-segmentation model (classes: whole stoma, pore) trained on manually annotated crops plus hard-negative background images was run tile-wise on full-resolution micrographs. For each stoma we measured length, width, guard-cell width and pore aperture; size statistics use only well-focused, non-edge stomata. Pore aperture is refined at sub-pixel resolution from intensity profiles across the pore. Density = all counted stomata / imaged leaf area.

Theoretical conductance follows Franks & Farquhar (2001): $g_{s,\max} = \frac{d}{v} \frac{D\, a_{\max}}{l + \frac{\pi}{2}\sqrt{a_{\max}/\pi}}$, with density D, pore area $a_{\max}$ (from pore length and aperture) and pore depth l approximated by guard-cell width.

Statistics. Each trait was analysed per micrograph with a linear mixed model, `trait ~ cytotype * leaf_side + (1 | population) + (1 | population:plant)`, accounting for repeated images per plant. Means are model-estimated marginal means (EMMs); pairwise contrasts are Tukey-adjusted within leaf side.

Results

Table 1: Model-estimated marginal means per cytotype and leaf side ($g_{s,\max}$ in mol m^{-2} s^{-1}).

	Abaxial			Adaxial		
Cytotype	Length (μm)	Density (mm^{-2})	$g_{s,\max}$	Length (μm)	Density (mm^{-2})	$g_{s,\max}$
ory_2x	18.5	296.9	1.03	19.0	243.0	0.84
nat_2x	18.7	254.4	0.91	19.1	208.3	0.74
nat_4x	20.6	236.9	0.92	21.4	211.7	0.84
neo_4x	23.9	164.2	0.76	24.6	138.0	0.64

Table 2: Selected pairwise differences (first minus second cytotype), Tukey-adjusted within leaf side. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, ns not significant.

	Abaxial			Adaxial		
Contrast	Length	Density	$g_{s,\max}$	Length	Density	$g_{s,\max}$
nat_4x - nat_2x	+1.9 ***	-17.4 ns	+0.00 ns	+2.3 ***	+3.5 ns	+0.10 ns
neo_4x - nat_2x	+5.1 ***	-90.2 ***	-0.15 ***	+5.5 ***	-70.3 ***	-0.09 *
neo_4x - nat_4x	+3.2 ***	-72.8 ***	-0.15 **	+3.2 ***	-73.8 ***	-0.20 ***
ory_2x - nat_2x	-0.2 ns	+42.5 ***	+0.12 **	-0.1 ns	+34.8 **	+0.11 **

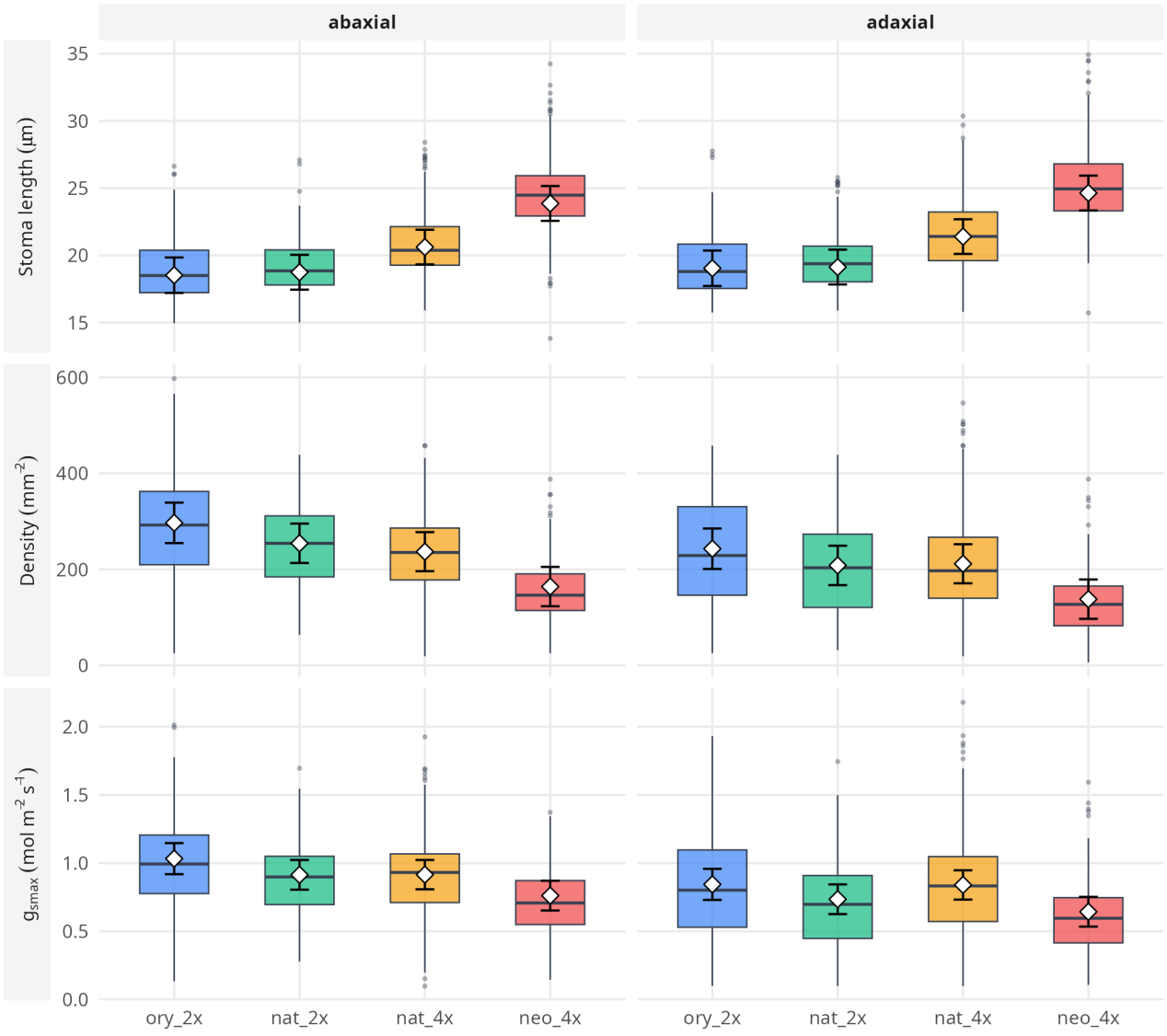


Figure 1: Stoma length, stomatal density and theoretical maximum conductance per micrograph. Boxes show image-level distributions; white diamonds with bars show model EMMs ± 95% CI.

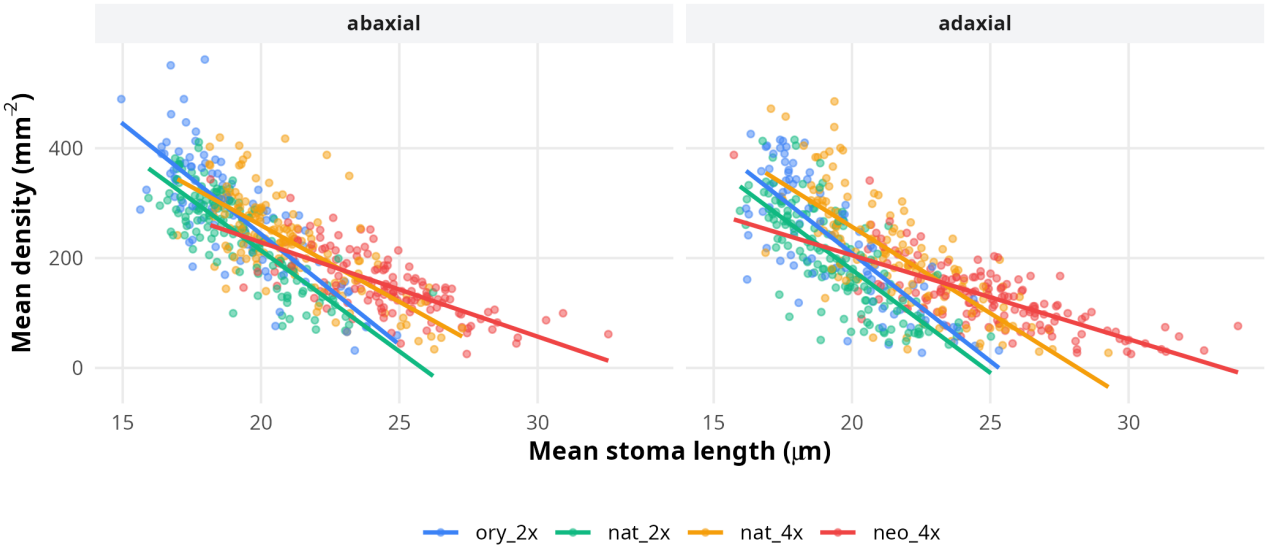


Figure 2: Size–density trade-off. Each point is a plant mean; lines are per-cytotype linear fits.

Comparison with the previous model

Results above come from the newly trained model. Below they are compared with the previous model's run on the same micrographs (previous run vs. new run). Across 3,253 matched images, per-image density correlates at r = 0.99 and median stoma length at r = 0.98; the stoma count is identical in 23% of images and within ±1 in 55%.

Table 3: Training metrics at the best epoch (validation split). The validation sets differ between dataset versions, so these numbers are only roughly comparable.

Metric	Previous	New
Epochs trained	120	120
Best epoch	117	120
Box precision	0.980	0.930
Box recall	0.961	0.916
Box mAP50	0.960	0.948
Box mAP50-95	0.700	0.723
Mask mAP50	0.960	0.972
Mask mAP50-95	0.704	0.671

Table 4: Detection summary across all micrographs, previous vs. new model.

Quantity	Previous	New
Micrographs	3,253	3,253
Stomata counted	100,679	105,354
Stomata per image (mean)	30.9	32.4
Images with no stomata	0	0
Stomata used for sizing	49,597	51,460
Median detection confidence	0.921	0.921
Detections with confidence < 0.45 (%)	5.5	5.5

Table 5: Abaxial EMMs from the previous and the new model (previous → new).

Cytotype	Length (μm)	Density (mm^{-2})	$g_{s,\max}$
ory_2x	18.3 → 18.5	283 → 297	–
nat_2x	18.6 → 18.7	242 → 254	–
nat_4x	20.3 → 20.6	226 → 237	–
neo_4x	23.5 → 23.9	157 → 164	–

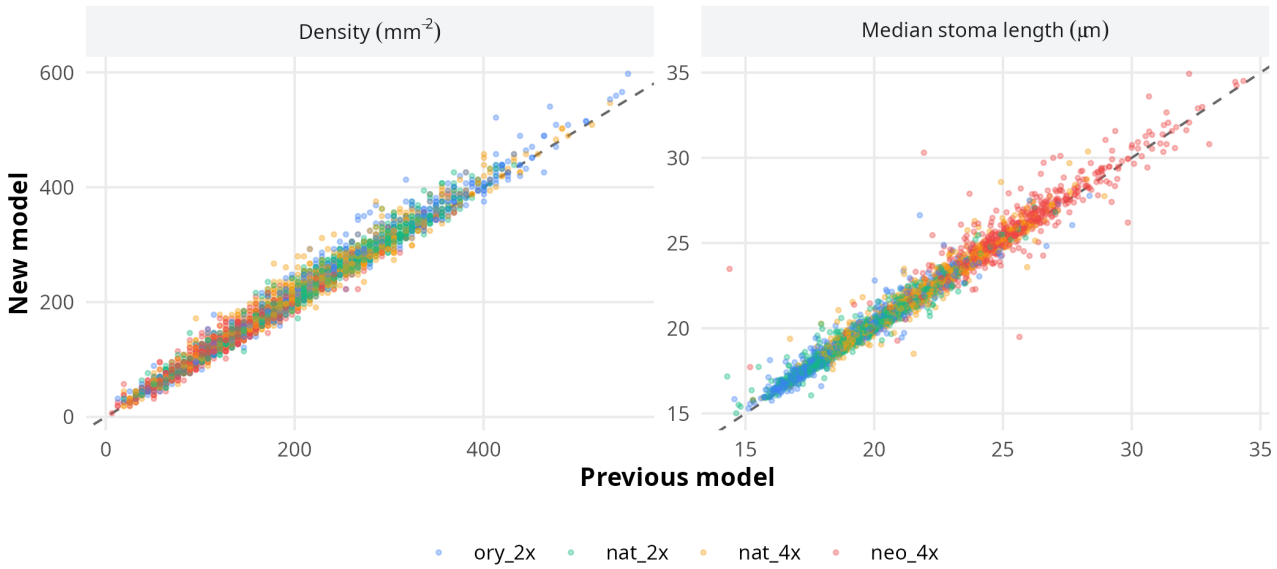


Figure 3: Per-image agreement between the previous and the new model. Dashed line: 1:1.

Caveats

- Results are **preliminary**; measurements come from an automated pipeline and have not yet been fully validated against manual counts on an independent set.
- $g_{s,\max}$ is a *theoretical* anatomical estimate using guard-cell width as a proxy for pore depth and the imaged (not maximal) aperture; it is best used for relative comparisons among cytotypes, not as absolute conductance.
- With only 5 populations, the population-level variance component is estimated imprecisely.